

Annaly Ellis

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EDUCATION

University of California, San Diego

BS in Data Science, specialization Machine Learning & Artificial Intelligence

La Jolla, CA

Sep. 2023 – Present

TECHNICAL SKILLS

Languages: Python, Java, R, JavaScript, HTML/CSS, MATLAB, SQL

Frameworks & Libraries: TensorFlow, Pytorch, scikit-learn, vLLM, pandas, NumPy, React, Matplotlib, Plotly, seaborn, Flask

Tools & Platforms: AWS (EC2), Git, PostgreSQL, CVAT, Google Cloud, Figma, VS Code, IntelliJ, RStudio

RELEVANT COURSEWORK

Data Structures and Algorithms, Deep Learning, Probabilistic Modeling & Machine Learning, Recommender Systems, Systems for Scalable Analytics, Data Visualization, Data Management, Product Management, Mathematical Statistics, Calculus, Linear Algebra

EXPERIENCE

Data Science Research Intern

Sep 2024 – Present

Scripps Institution of Oceanography

La Jolla, CA

- Improved glacier calving detection from 10% to 93% accuracy using TensorFlow and PyTorch CNN+LSTM architectures.
- Trained models on time-series satellite imagery across 100+ Sentinel-2 tiles of Greenland (110km x 110km each).
- Built scalable Python pipelines to preprocess optical and radar geospatial datasets across all of Greenland.
- Contributed to a seismology research paper by linking global seismic events to calving regions via statistical filtering.

Machine Learning Engineer Intern

Jan 2026 – Mar 2026

Veeva Health

Remote

- Supported development of a deep learning model predicting coronary artery disease and plaque from medical imaging data.
- Produced 3D visualizations of coronary artery structures to aid team review and interpretation of model outputs.
- Researched CT, MRI, and other major imaging modalities used in ML to inform preprocessing and model design decisions.

Software Engineering & Product Management Intern

Aug 2025 – Dec 2025

Automatiq AI

San Diego, CA

- Contributed to a full platform launch by leading front-end design of a micro-SaaS product with a team of 5+ engineers.
- Built front-end features for AI marketing platform that generates strategy, content, ads, and funnels using multiple LLMs.
- Improved usability by refining UI/UX components for intuitiveness and accessibility, created user-friendly designs.

Computer Science Mentor

July 2025 – Sep 2025

UCSD Department of Computer Science and Engineering

La Jolla, CA

- Taught 50+ UCSD students about Python, data structures, algorithms, OOP, dynamic programming, and machine learning.
- Supported growth and improved student success rate in ~30 full-stack ML & web development final projects.
- Led lab hours and review sessions helping students debug code and solve challenging technical problem sets.
- Strengthened program community as Social Lead by organizing events that increased collaboration across students and staff.

PROJECTS

SheSteps - ML Powered Women's Safety | CalHacks 2025

Oct 2025

- Delivered a women's safety mobile app in a 3-person team by leading design and building a scikit-learn ML pipeline.
- Tuned model to assess risk levels and optimize running/walking routes across 930 sq. miles of real-time SF crime data.
- Integrated a Flask REST API with a React Native + TypeScript frontend to serve real-time route recommendations.
- Designed UI/UX in Figma with Google Maps and LiveKit Voice AI Agents for accessible, high-stress navigation scenarios.

Emergency Medical Services AI Integration | Helpert AI Product Competition

May 2025 – June 2025

- Won 2nd place overall and \$2,000 in a judged AI product competition with a panel of 10 judges.
- Led design of an AI assistant for EMTs supporting real-time guidance, auto-documentation, and hospital communication.
- Researched EMS workflows to define product scope and pricing strategy, leading to hospital board deployment discussion.
- Trained the AI system for intelligent prompting, speech navigation, labeling, and summary generation based on live data.

Environmental Sustainability Web App | UCSD Women in Computing

Sep 2024 – Nov 2024

- Collaborated with 5 engineers in an agile environment to build a climate change education focused web app.
- Designed an interactive JavaScript quiz and responsive web pages using HTML and CSS for cross-browser compatibility.